

4.6 Lymphocytes

▼ What is the difference between innate and adaptive immune cells?

- **Innate** immune cells are involved in **engulfing and killing pathogens** this way (i.e macrophages, neutrophils)
- **Adaptive** immune cells are involved in **targeting specific regions** on pathogens (i.e T cells, B cells)

▼ What are the main functions of T cells?

- To **orchestrate an immune response** (T helper) and **kill infected cells** (T killer)

▼ What are the main functions of B cells?

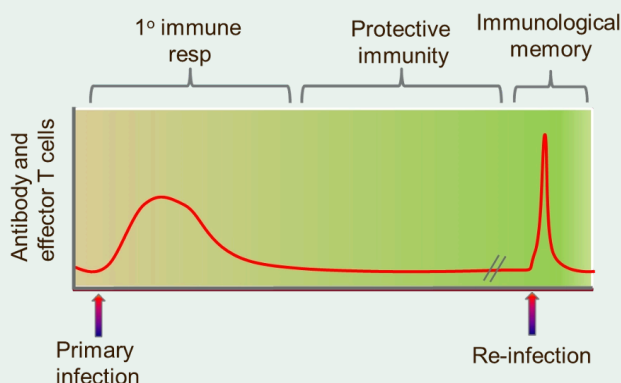
- To **make antibodies**

▼ Why is adaptive immunity important?

- To clear initial infections and **protect us from repeated exposure to these pathogens** via memory

▼ Why are memory responses typically faster than primary responses?

- The process for a naive cell to become an effector cell is one that requires a lot of signalling
- But the process for a memory cell to respond to a familiar pathogen is quite quick and doesn't require much signalling



▼ How long can immune memory last for?

- A very, very long time - can be **lifelong**
- There are cases which suggest to us that it could last for up to 60 years.

▼ What are the two types of T cell?

- CD4
- CD8

▼ What do CD4 T cells do?

- They **produce cytokines** to help **shape the immune response**.

▼ What do CD8 T cells do?

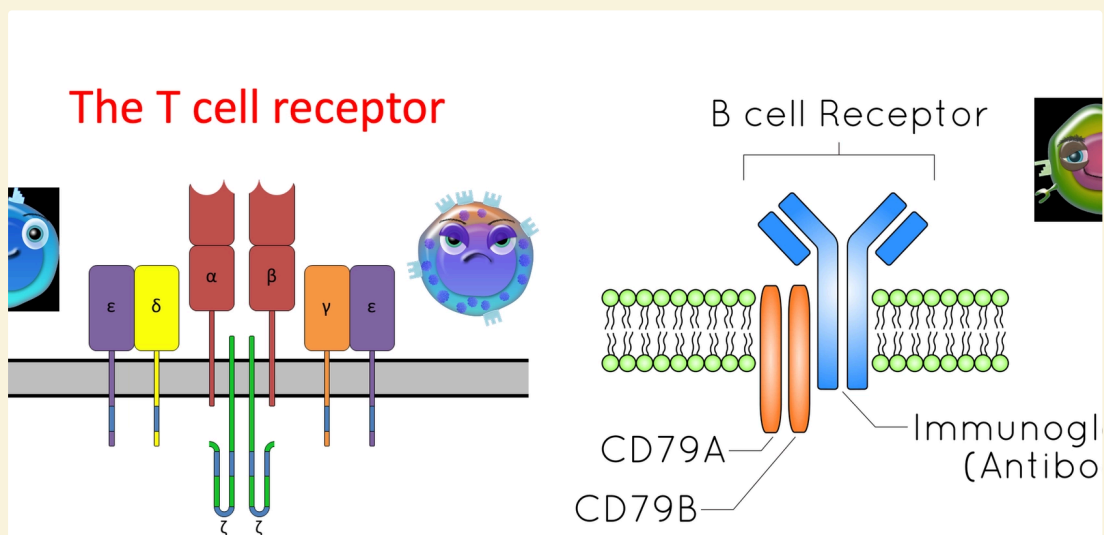
- They **kill infected cells**

▼ What part of a T or B cell allows it to interact with antigens?

- The **T or B cell receptor**.

▼ How is a B or T cell activated?

- The **antigen binds onto the receptor**
- This causes the **proteins under the receptor region to bind onto other proteins** in a signalling cascade



▼ What is an epitope?

- The **region on an antigen where a B or T cell receptor binds to**

▼ How do T cells bind to antigens?

- **T cells** recognise a linear sequence of ~10-15 amino acids on the epitope on an **MHC (major histocompatibility complex)**
- They recognise the **primary structure** in the epitope

▼ How do B cells bind to antigens?

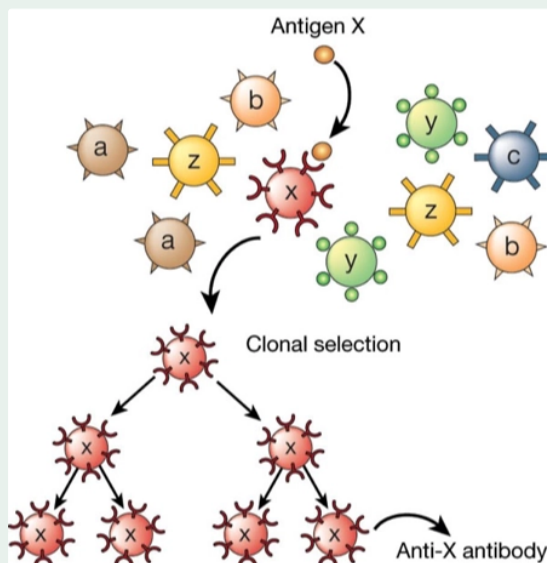
- They recognise the **structure/shape of the epitope** rather than the sequence
- They recognise the **tertiary/quaternary structure of the epitope**

▼ What is a cognate antigen?

- The antigen that a specific T cell recognises

▼ Describe what clonal expansion is.

- When a vaccine is given for example, there are small amounts of T cell and B cell produced in response to the spike proteins
- Clonal expansion describes how the number of these T and B cells increase as they are re-exposed to their cognate antigen as they have to produce an immune response.



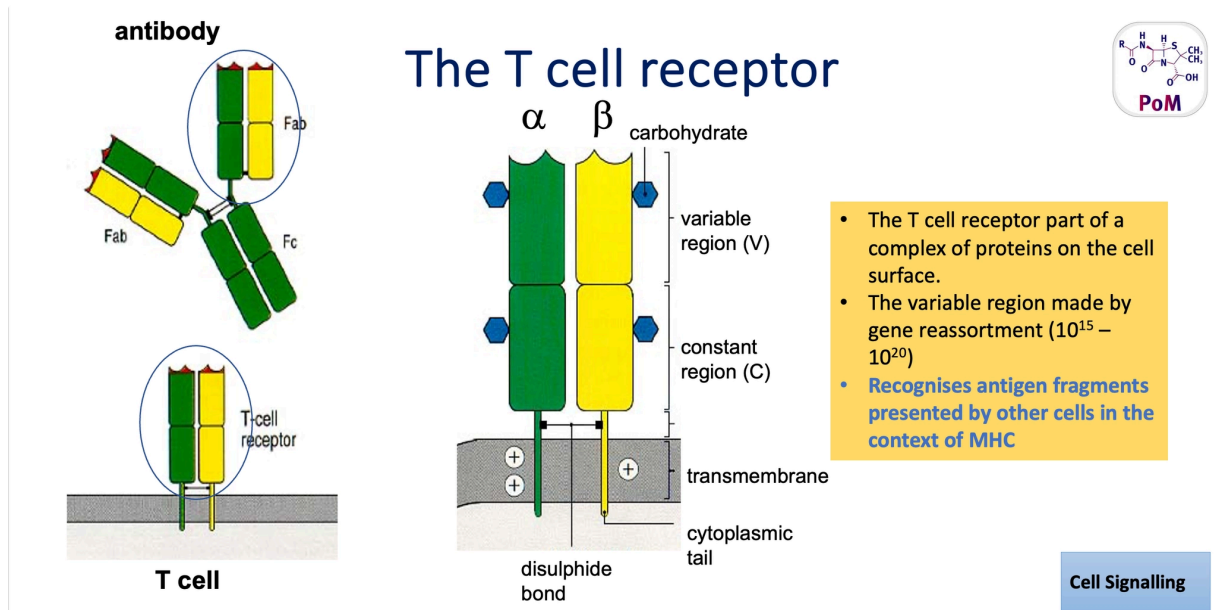
▼ What is immunoglobulin gene rearrangement and why is it important?

- It is the process where the genes are rearranged during protein production in order to **increase the B and T cell receptor diversity**
- This is because there are **not enough genes** to combat the **large number of pathogens** that exist globally and so this method of **rearrangement helps** to increase the number of receptors possible by rearranging genes

▼ What happens to T cell production as we age?

- It **decreases**.
- Once we hit the **age of ~ 25-26**, **T cell production in the thymus stops** so we stop making new T cells and our **T cell repertoire doesn't increase in size**.

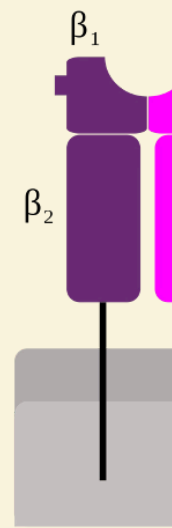
- ▼ Diagram showing the structure of a T cell receptor.



- ▼ What is the role of the major histocompatibility complex (MHC)?
- To **present antigens on cell surfaces**, so that T cells can interact with them and initiate a response.
 - It also plays a key role in **defining self and non-self** antigens
- ▼ What is the role of the HLA gene family?
- To **code for the major histocompatibility complex (MHC)**

▼ Describe three differences between MHC Class I and Class II.*

- **MHC Class I** is found in **all nucleated cells**, but **MHC Class II** is **only found** in professional **antigen-presenting cells**
- **MHC Class I** is used against **intracellular pathogens**, **MHC Class II** is used against **extracellular pathogens**
- **MHC Class I** has a **single variable alpha chain** and a **common beta-microglobulin**. **MHC Class II** has an **alpha and a beta chain**



▼ What term is used to describe MHC gene expression?

- Polygenic.
- It has genes that code for it at different loci.

▼ Which T cells does MHC Class I interact with?

- CD8

▼ Which T cells does MHC Class II interact with?

- CD4

▼ Which cells are used to produce cytokines?

- T helper cells

▼ How are the different classes of T helper cells useful?

- They are used to provide **different types of responses** (i.e Th1, Th2, Th17 response) **to pathogens**

▼ When are cytokines produced?

- When a T helper cell is responding to a pathogen

▼ What is the role of cytokines?

- To act as inflammatory mediators

▼ How do T killer cells (CD8) recognise infected cells?

- Through the recognition of the **MHC Class I and antigenic proteins**

▼ Describe the mechanism used by CD8 cells to kill infected cells.

- They have a **core protein** called **perforin (hydrophilic)** which **insert a hole into host cell membrane** that **allows hydrophilic molecules** to go through
 - Previously hydrophilic molecules couldn't get through due to the hydrophobic nature of the lipid membrane
- This allows the **CTL (cytotoxic T-lymphocyte/T cell)** to **insert granzyme** into the cell which **triggers a cascade of reactions leading to apoptosis**

▼ What is the name of the other pathway used by CD8 cells to kill infected cells?

- Fas and Fas-ligand pathway

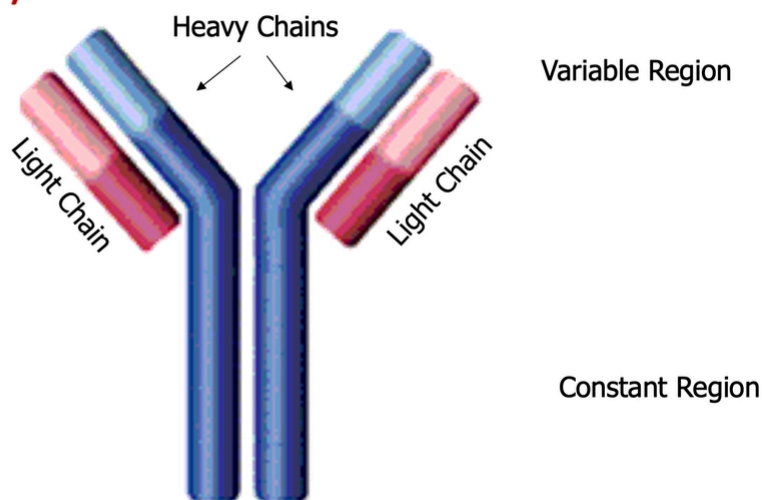
▼ Why are B cells said to be part of the humoral immune response?

- Because they secrete antibodies
- The **humoral immune response** describes the **role of antibodies** in providing immunity

▼ Describe the structure of an antibody.

- Constant region (bottom) and a variable region (top)
- Four chains, two light (short) and two heavy (long)

Antibody structure

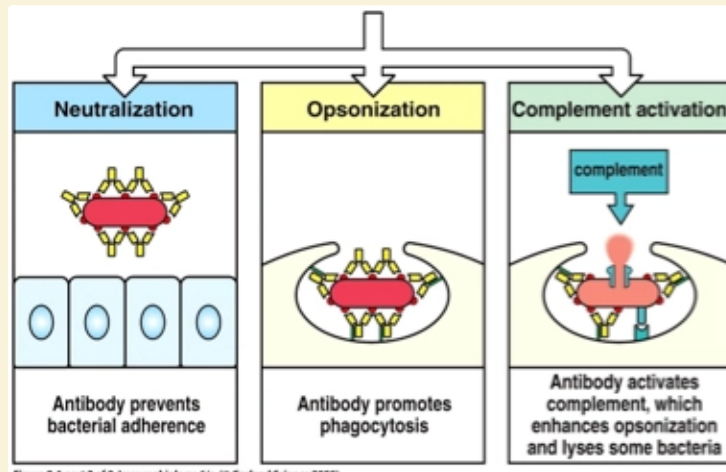


▼ What is the role of the variable region on antibodies?

- To allow the antibody to bind to different types of antigen.

▼ What are the three methods used by antibodies to protect against infections?

- Neutralisation
- Opsonisation
- Complement activation



▼ Describe how the process of neutralisation works.

- Neutralisation is when an **antibody prevents a virus from binding onto a host cell receptor**
- It does this by **binding onto a region of the virus called the receptor binding domain** and so the **virus is no longer able to bind onto host cells** and use their machinery for proliferation.

▼ Describe how the process of opsonisation works.

- This describes how **covering a pathogen with antibodies increases the likelihood of an innate effector cell** (those that phagocytose) **phagocytosing the pathogen**

▼ What are the roles/characteristics of IgE?

- Fighting against **allergy**.

▼ What are the roles/characteristics of IgG?

- Important in the serum
- **Highest neutralisation and opsonisation activity.**
- It has **four major subclasses** in humans.

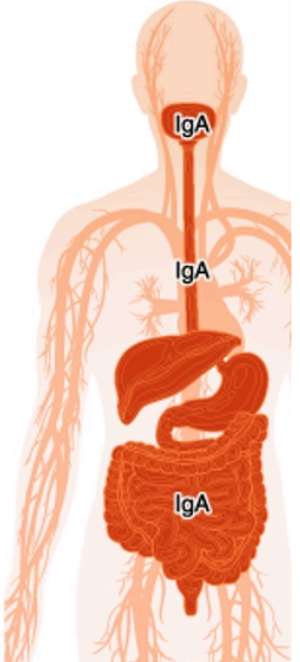
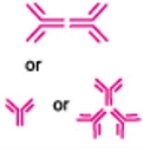
▼ What are the roles/characteristics of IgM?

- **Produced upon initial antigen invasion** and increases transiently
- Forms a **pentamer**

▼ What are the roles/characteristics of IgA?

- It is expressed in **mucosal tissues**
- It **forms dimers after secretion**.

▼ Table showing the role of different antibody types.

Types and characteristics of antibodies		Distribution in the body
	• Highest opsonization and neutralization activities. • Classified into four subclasses (IgG1, IgG2, IgG3, and IgG4).	
	• Produced first upon antigen invasion. Increases transiently.	
	• Expressed in mucosal tissues. Forms dimers after secretion.	
	• Unknown function.	
	• Involved in allergy.	

▼ What is the difference between the way B and T cells bind to antigens?

- B cells bind to soluble antigens roaming freely in the blood
- T cells bind to antigens on the surface of MHC complexes on APCs

▼ What three things need to happen for a B cell to be activated?

- Recognise the antigen that it's going to make antibodies against
- Have surface contact with a T cell (co-stimulatory contact)
- Become activated by **cytokines** (released by CD4 T cells)

▼ What are thymus-independent antigens?

- These are antigens, (often polysaccharides) that can **activate B cells without the help of T cells**
- **Signal from the pathogen (PAMP) is enough to activate the B cell.**

▼ Which antibodies are activated by thymus-independent antigens?

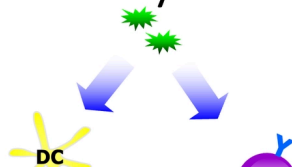
- **IgM**

▼ Describe how B cells are activated by T cells.

- A **B cell that has antibody on its surface** is able to recognise a specific antigen
- A **dendritic cell** is also able to recognise this specific antigen
- **Both of these cells bind onto the pathogen in question, and present the antigen on their MHC II complexes**
 - This allows the same T helper cell to recognise both the dendritic cell and the B cell it needs to communicate with
- The **T helper cell binds onto the antigen on the dendritic cell surface**, and tells the B cell to proliferate and differentiate into **plasma cells which create and secrete more and more antibody against the antigen in question**
 - The T helper cells also signals to T killer cells, so they can **kill infected cells**

▼ Diagram showing how B cells can be activated by T cells "training them" to make antibodies.

} cell activation by T cells



- 1) The membrane bound BCR **recognises** antigen
- 2) The receptor-bound antigen is **internalised** and **degraded** into peptides
- 3) Peptides associate with "self" molecules (MHC class II) and is **expressed** at the cell surface